Analytics in Managerial Economics

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About me

- My path:

- My research:
  Risk management, decision making under uncertainty, banking regulation, and information economics

- I have consulted several startups, Fortune 100 companies, and financial institutions.
  Almost all about analytics…

About you

- What is your experience with economics or data science?

- Any data analytics opportunities or threats you are aware of?

This course is NOT about how to use typical machine learning models

- So, this is not an advanced optimization, machine learning, or statistics course
- However, we teach causal inference!

Example model

- You have the following regression model:

$$Y_t = a + bX_t + e_t,$$

where Y_t is a dependent variable (outcome variable), X_t is an independent variable (explanatory variable), e_t is an error term, a and b are regression parameters.

- After estimating the parameters, you have:

$$Y_t = 0.2 + 0.5X_t + e_t,$$

where now both Y_t and X_t are binary, so 0 or 1.

- Question:
  If you can change X_t and if you prefer Y_t to be equal to 0, what should you do? And why?

Correlation is not causality

Would there be less accidents if we removed the warning signs?

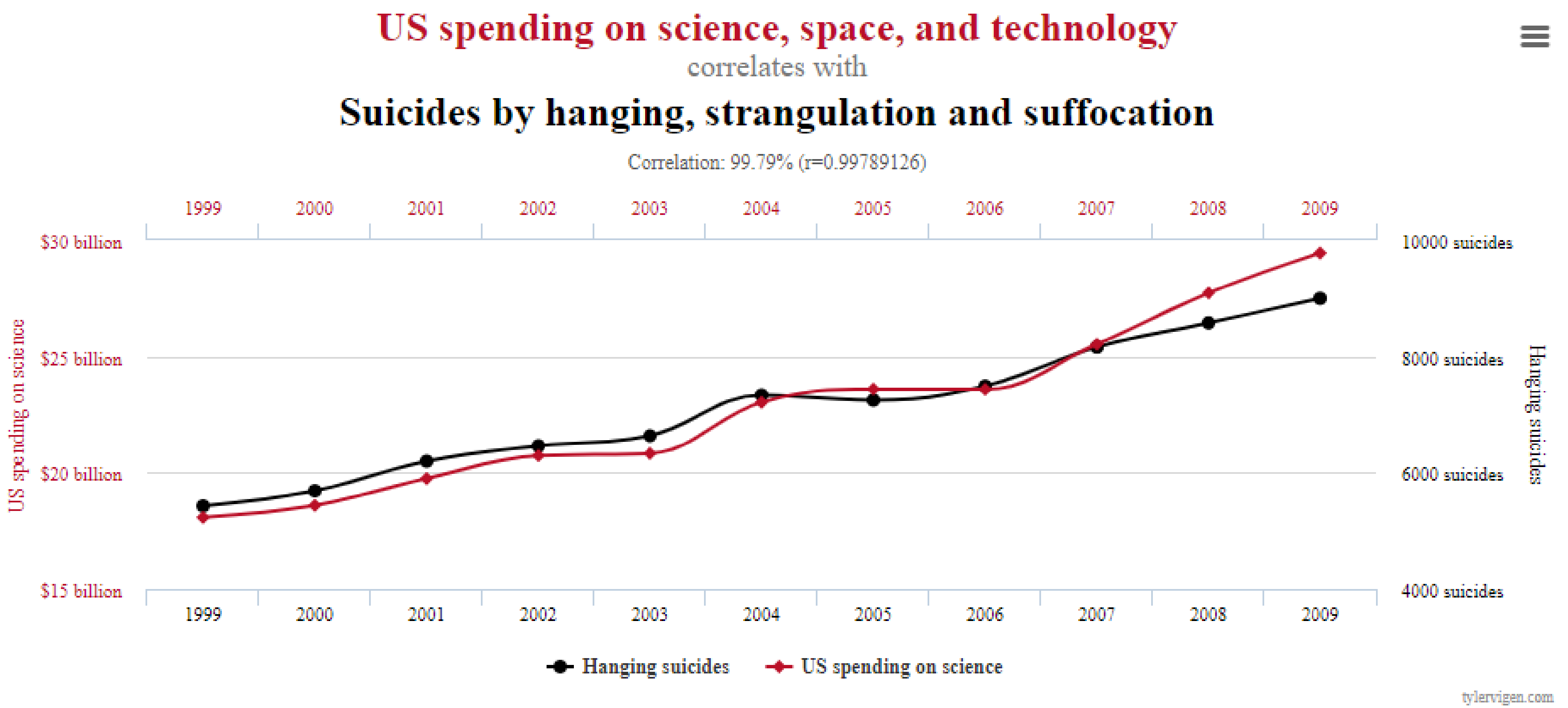
Correlation is not causality (Cont'd)

What is happening here?

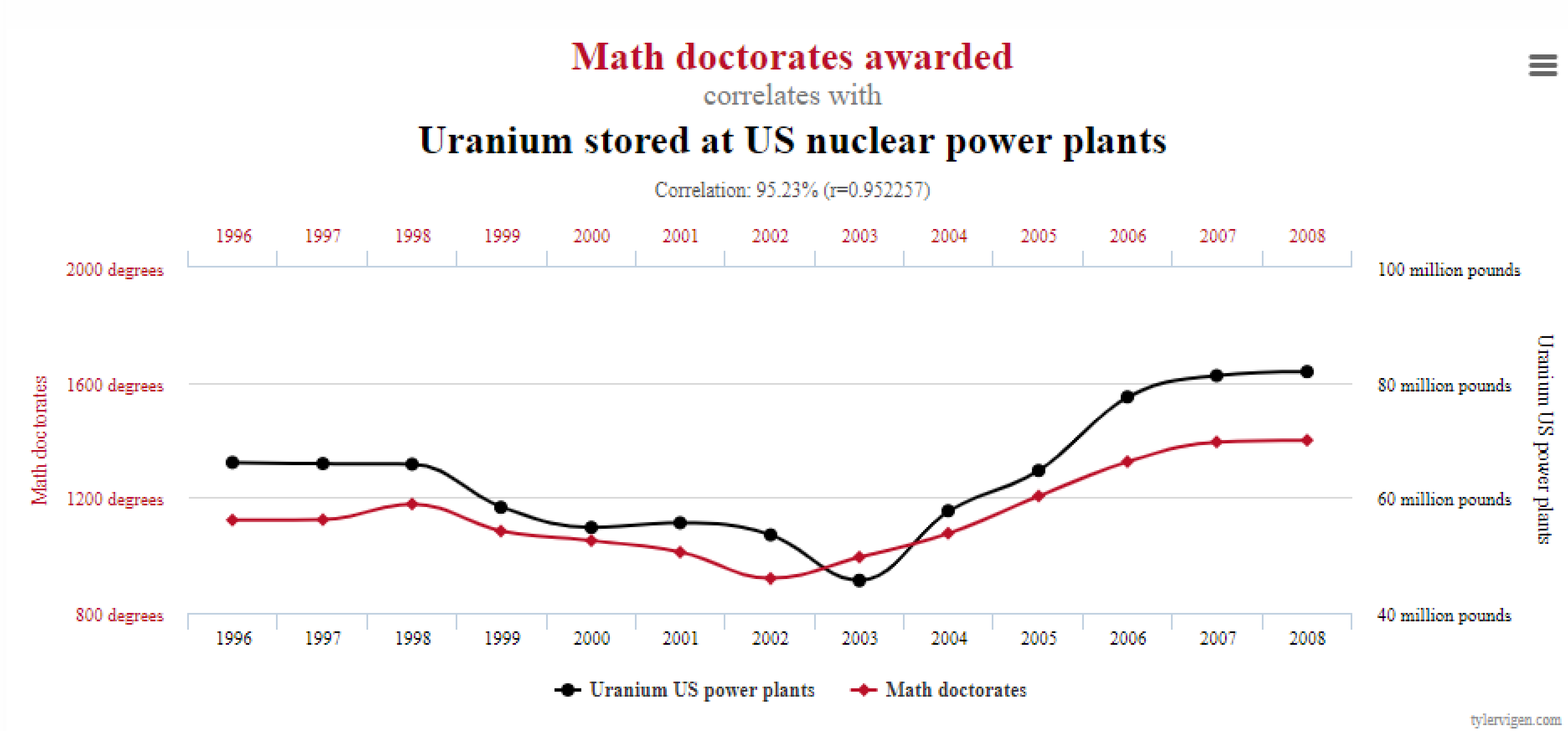
Wet floor

Warning sign

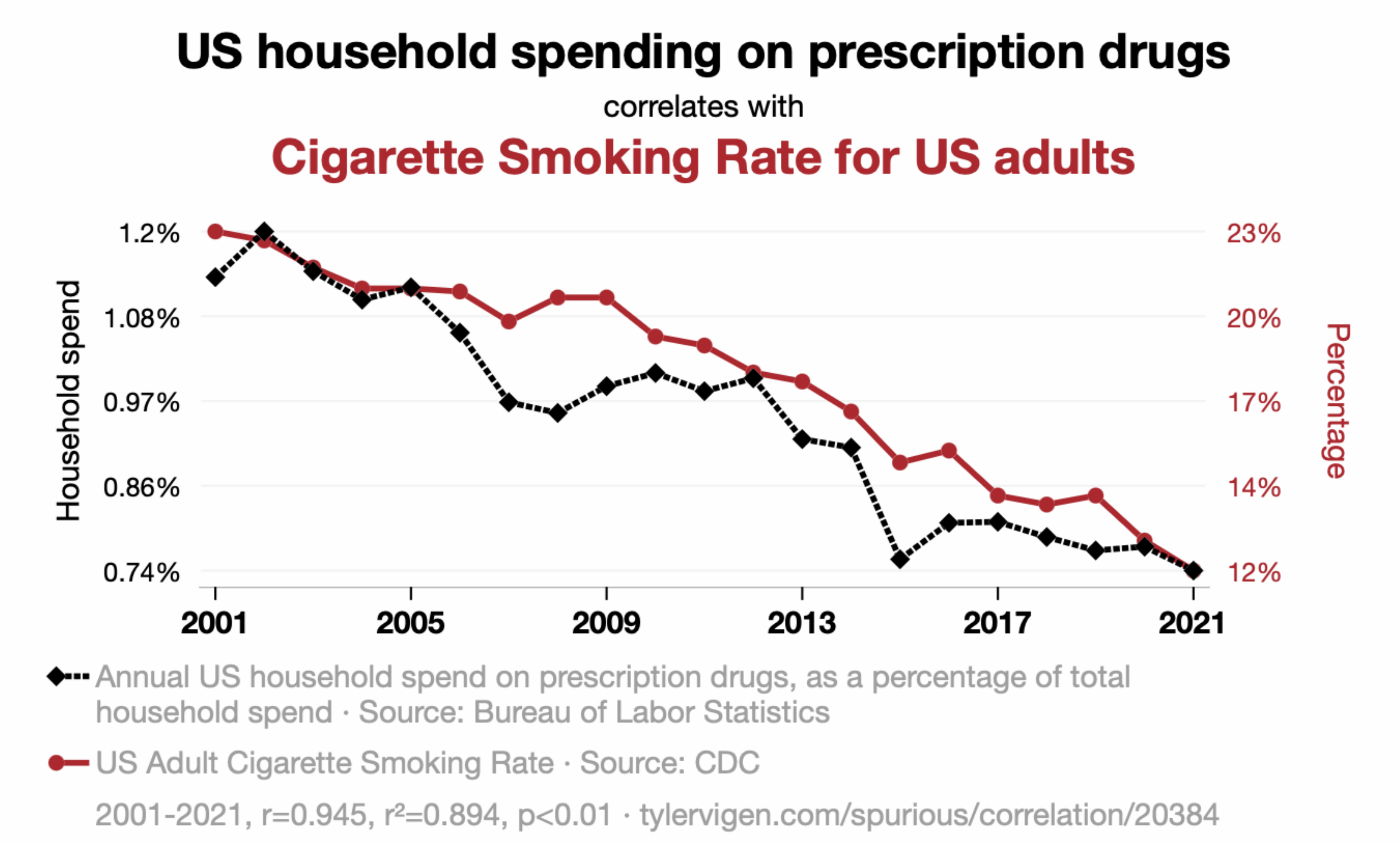
Accidents



Data sources: U.S. Office of Management and Budget and Centers for Disease Control & Prevention



Data sources: National Science Foundation and Dept. of Energy


US household spending on prescription drugs
correlates with
Cigarette Smoking Rate for US adults
1.2%
1.08%
0.97%
0.86%
0.74%
Household spend
23%
20%
17%
14%
12%
Percentage
2001
2005
2009
2013
2017
2021
Annual US household spend on prescription drugs, as a percentage of total household spend · Source: Bureau of Labor Statistics
US Adult Cigarette Smoking Rate · Source: CDC
2001-2021, r=0.945, r²=0.894, p<0.01 · tylervigen.com/spurious/correlation/20384

US household spending on prescription drugs also correlates with...

Variable	Correlation	Years	Sys. Score
The divorce rate in North Carolina	r=0.95	22yrs	412
Popularity of the first name Trey	r=0.95	23yrs	390
The number of floral designers in Pennsylvania	r=0.96	20yrs	390
The number of janitors and cleaners in West Virginia	r=0.96	20yrs	390
US average milk-fat content of milk fat and skim solids byproduct fluid beverage milk	r=0.95	22yrs	388
The number of file clerks in Alabama	r=0.95	19yrs	388
The divorce rate in Washington	r=0.95	22yrs	382
The divorce rate in Kentucky	r=0.94	22yrs	380
Nuclear power generation in Germany	r=0.93	22yrs	379
Fossil fuel use in Denmark	r=0.92	22yrs	378
Kerosene used globally	r=0.94	22yrs	378
Physical album shipment volume in the United States	r=0.91	23yrs	376
U.S. intercountry adoptions	r=0.91	22yrs	374
Google searches for 'oprah winfrey'	r=0.92	19yrs	374
Milk consumption	r=0.94	22yrs	368
Burglaries in New York	r=0.94	23yrs	368
GMO use in cotton	r=0.94	23yrs	363
Cottage cheese consumption	r=0.91	22yrs	362
The distance between Neptune and the Sun	r=0.94	23yrs	362
The distance between Neptune and Earth	r=0.94	23yrs	362
The distance between Neptune and the moon	r=0.94	23yrs	362
Popularity of the first name Tanya	r=0.95	23yrs	360
Popularity of the first name Cayla	r=0.95	23yrs	360

What this course is about

- We learn how individuals and firms make decisions
  - Allocation of scarce resources
  - Interactions among these individuals and firms
- We study how to operate in different market conditions
  - For instance, we learn how to optimize prices for your products and services in these environments
- We estimate the models with different datasets and identification strategies
- Identify new opportunities and value in analytics
  - Many analytics opportunities are Economics 101 problems

Team

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Tentative schedule

There are two types of lectures: Economics theory and empirical methods. In the theory lectures, we apply the estimated effects from the empirical methods lectures to business problems, primarily focusing on pricing issues.

Lecture 1: Introduction, prediction vs causation
Lecture 2: Perfect competition and monopoly (*theory*)
Lecture 3: Linear regression, interaction variables, endogeneity (*empirical method*)
Lecture 4: Instrumental variables, two stage least squares (*empirical method*)
Lecture 5: Randomized controlled trials (*empirical method*)
Lecture 6: Difference in differences (*empirical method*)
Lecture 7: Third degree price discrimination (*theory*)
Lecture 8: First degree price discrimination (*theory*)
Lecture 9: Second degree price discrimination (*theory*)
Lecture 10: Multi-armed bandits (*empirical method*)
Lecture 11: In class project

We will have two lectures on Saturday, September 6 from 9am to 4pm. We will have a special session on LLM agents and their application to the course. The classroom will be announced later. Because of this, we will not have lectures during October 20 – 28.

Course Material

- Lecture notes/slides
- Angrist, Joshua D., and Jörn-Steffen Pischke. Mastering 'Metrics: The Path from Cause to Effect. Princeton University Press, 2015.
- Angrist, Joshua D., and Jörn-Steffen Pischke. Mostly harmless econometrics: An empiricist's companion. Princeton University Press, 2008.
- Cunningham, Scott. Causal Inference: The Mixtape. Yale University Press, 2021. Available at: https://mixtape.scunning.com
- Jeffrey M. Wooldridge: Introductory Econometrics, A Modern Approach, 5th Edition
- Lynne Pepall, Dan Richards and George Norman: Industrial Organization: Contemporary Theory and Empirical Applications, 5th Edition, Wiley
- Hal R. Varian: Intermediate Microeconomics, 8th Edition
- Daron Acemoglu, David Laibson and John List: Microeconomics, 2nd Edition, Pearson
- Slivkins, Aleksandrs: "Introduction to multi-armed bandits," 2021, available at https://arxiv.org/abs/1904.07272

Assessments

Component	Weight
In-class project	30%
Group projects	40%
Homework	30%
Exceptional participation gives up to 5% to the final grading.	
Total	**100%**

Projects

There are 3 case projects that are done in groups. Each group has 3 - 5 students. The due dates of the case projects: 20 September, 18 October, and 16 November.

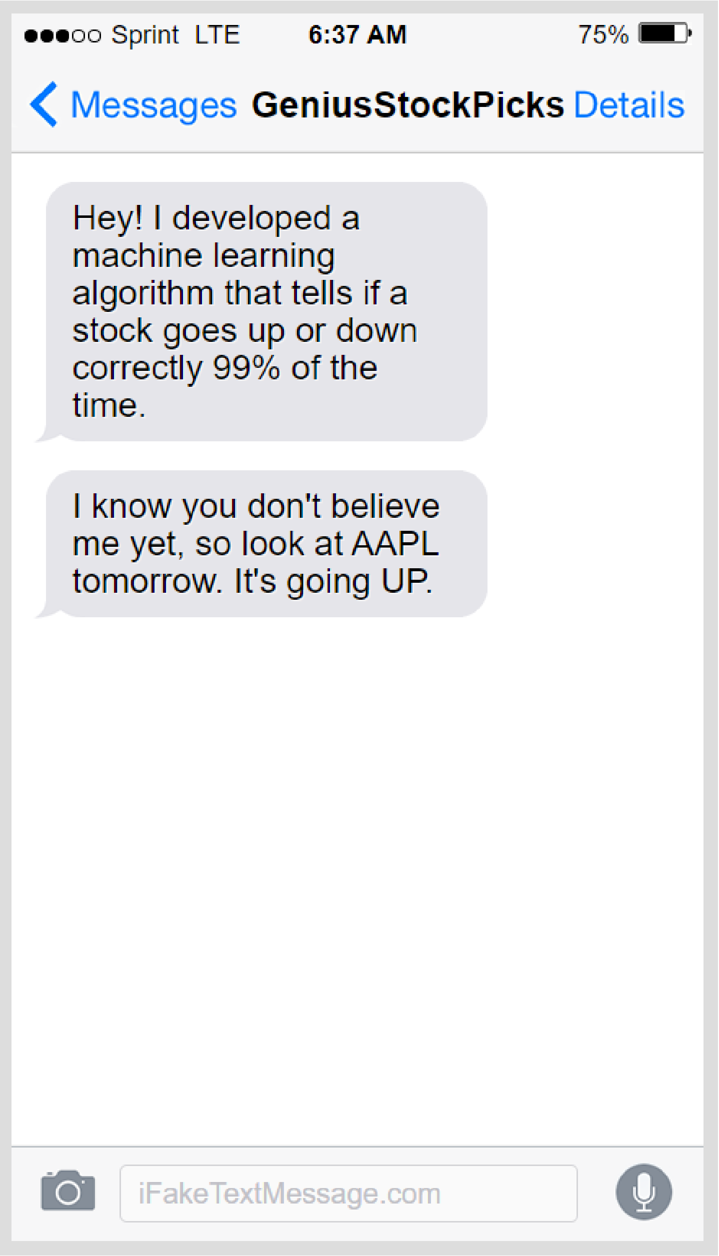
There is also one in-class individual project, which is on 11 November at 6pm. Furthermore, in addition to the projects there are homework assignments.

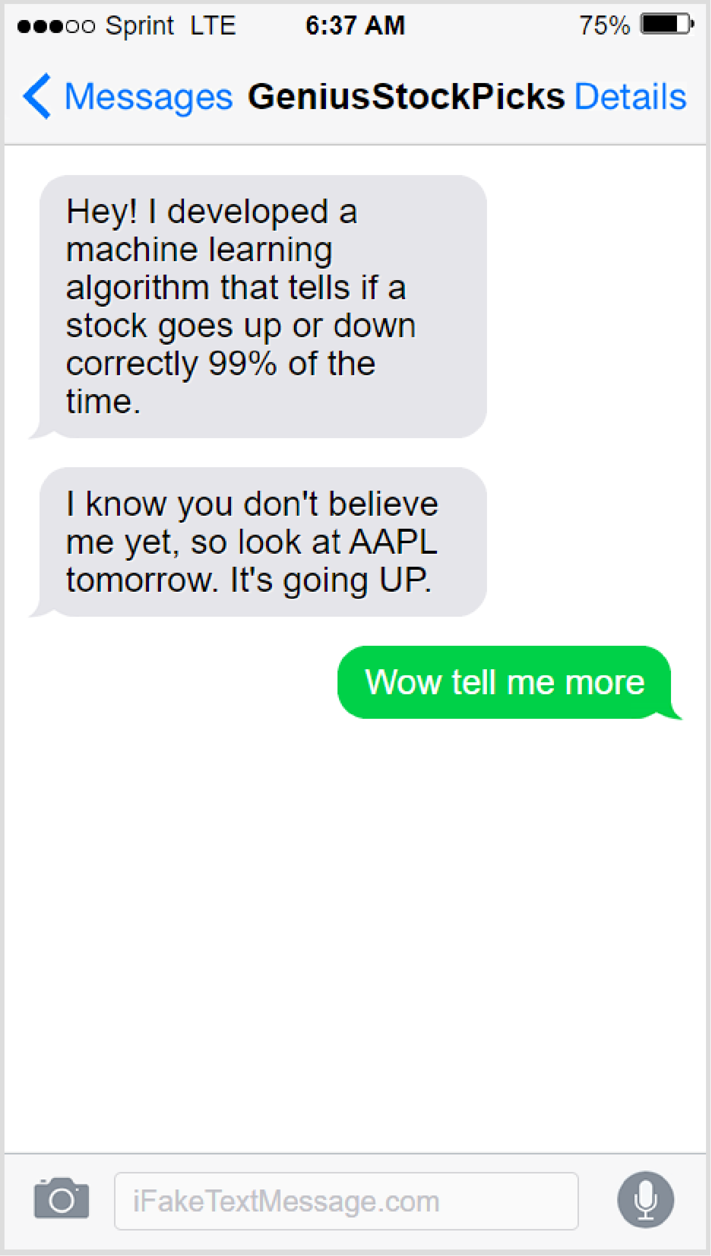
Housekeeping rules…

We have students with different backgrounds and experiences

- Your active participation is required to get your analytics problems and experiences discussed in the class

There are several homework and group assignments that needs to be studied

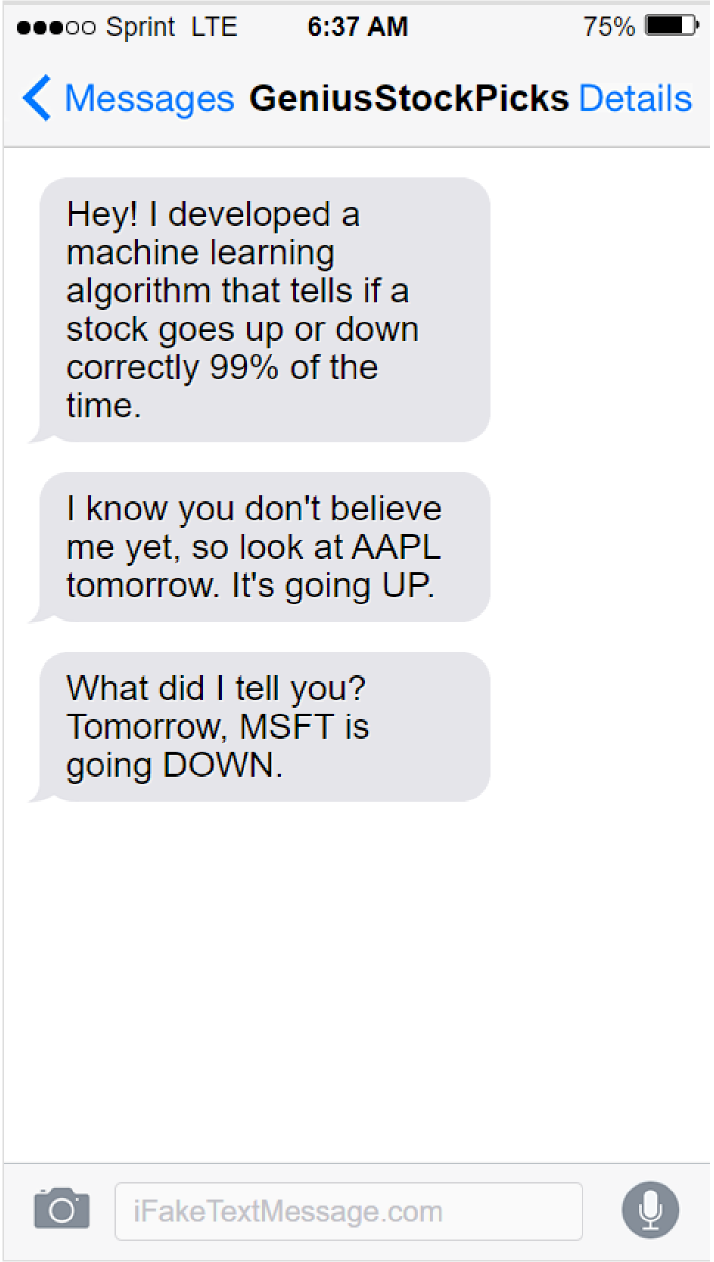

●●●○○ Sprint LTE
6:37 AM
75%
Messages
GeniusStockPicks
Details
Hey! I developed a machine learning algorithm that tells if a stock goes up or down correctly 99% of the time.
I know you don't believe me yet, so look at AAPL tomorrow. It's going UP.
iFakeTextMessage.com



●●●○○ Sprint LTE 6:37 AM 75%
Messages GeniusStockPicks Details
Hey! I developed a machine learning algorithm that tells if a stock goes up or down correctly 99% of the time.
I know you don't believe me yet, so look at AAPL tomorrow. It's going UP.
Wow tell me more
iFakeTextMessage.com

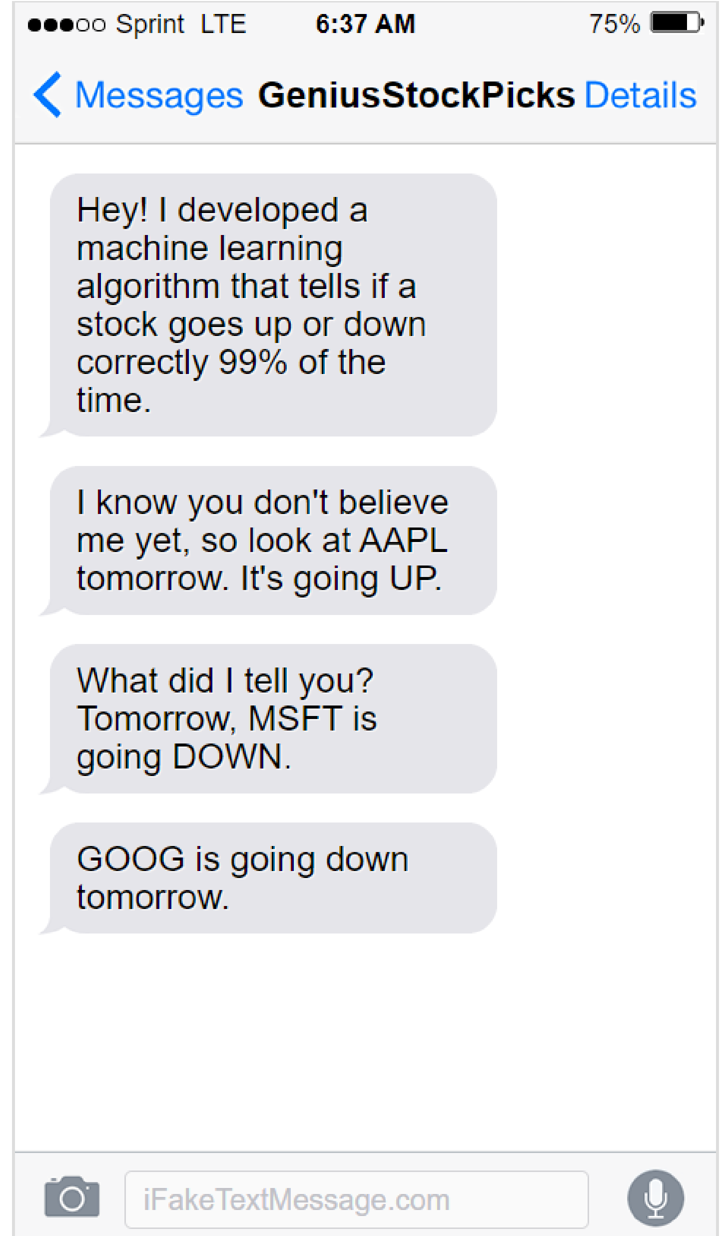
Next Day..

AAPL



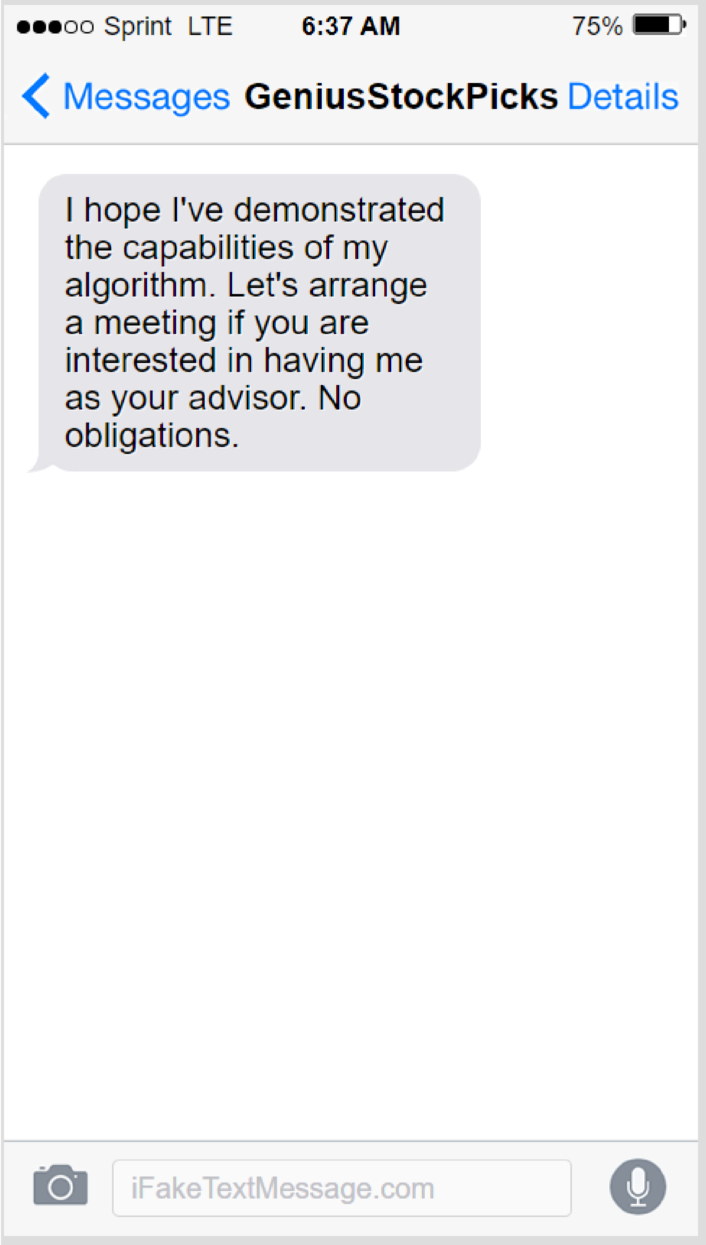
Next Day..

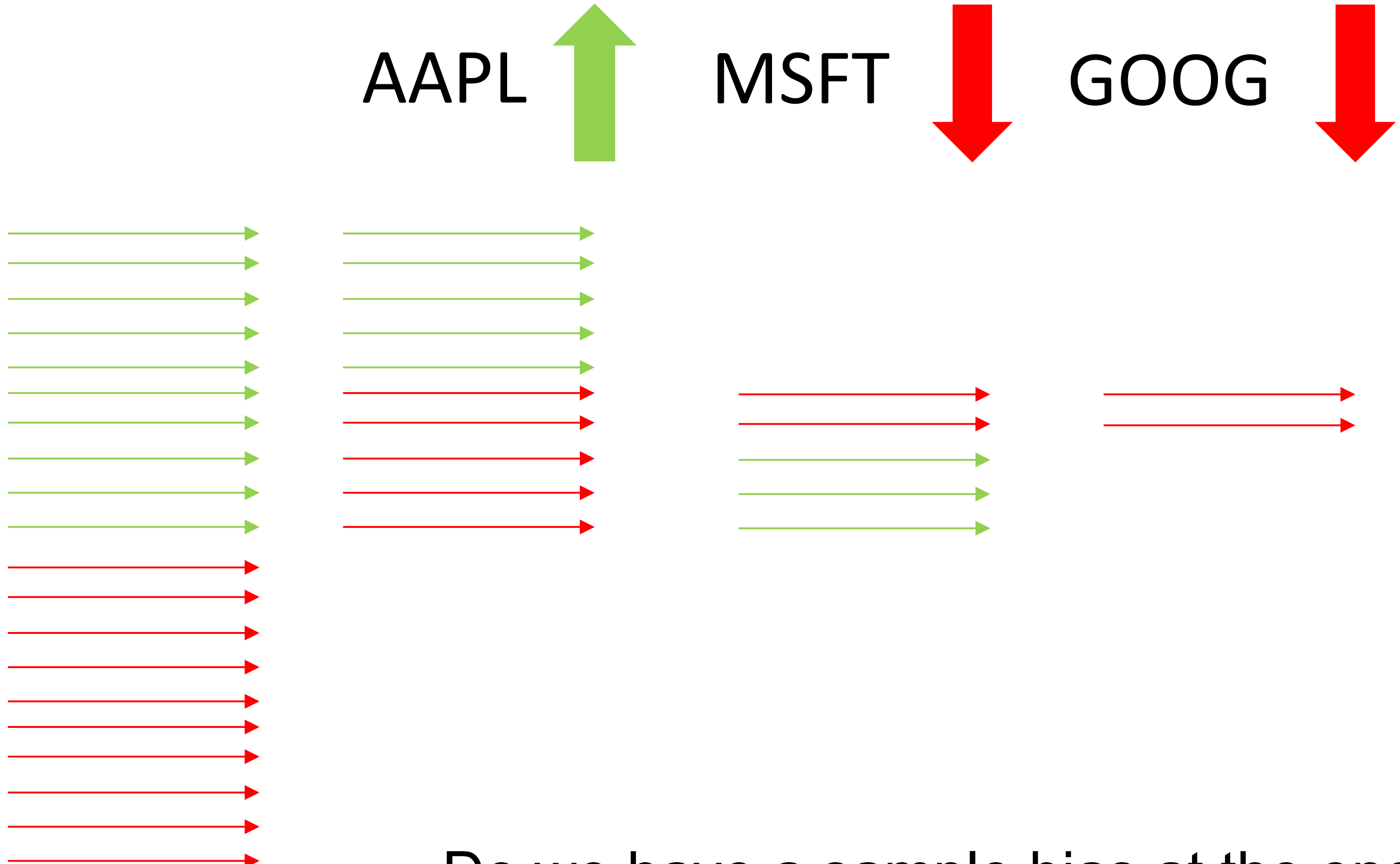
MSFT



Next Day..

GOOG





Do we have a sample bias at the end?

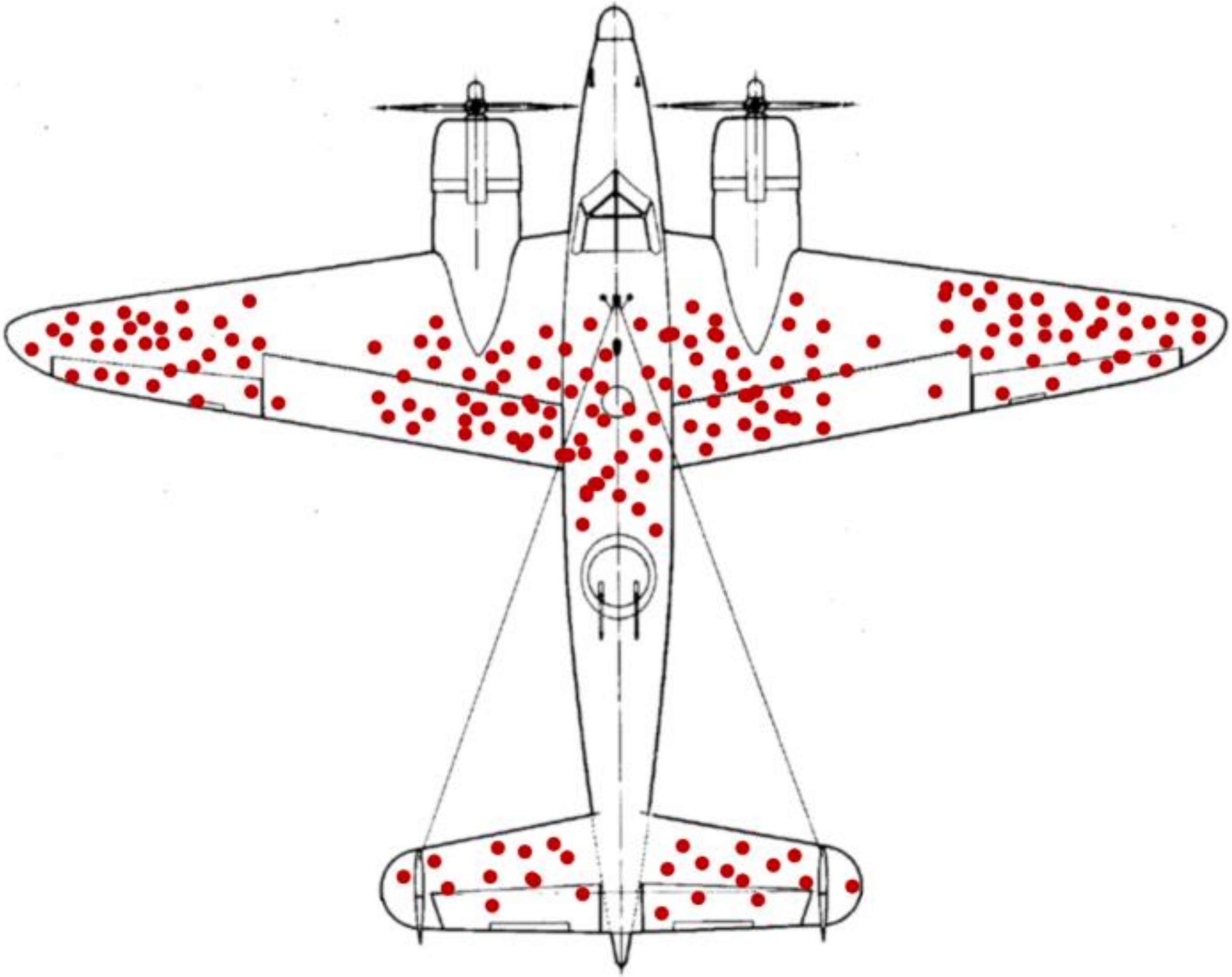
A healthcare clinic promotes a new wellness supplement to a large group of patients.

Claim: Supplement improves overall health and reduces common ailments.

First Round Filtering:
- After some time, only patients reporting health improvements are targeted again.
- The focus narrows to those who perceived benefits.

Subsequent Rounds:
- Repeated targeting of those reporting positive outcomes.
- Creates the impression of consistent, remarkable health improvements.

Survivorship Bias:
- Patients experiencing positive outcomes believe the supplement is highly effective.
- Many others, who saw no benefit, are ignored, skewing the perception of efficacy.



Survivorship bias

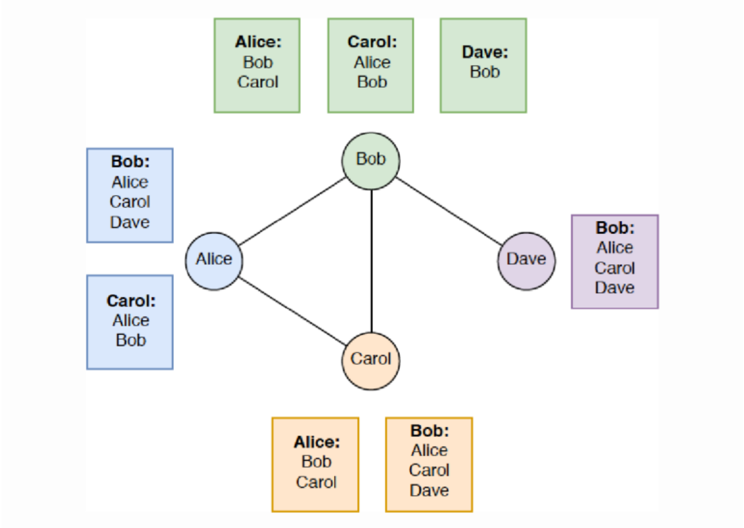
- Not looking at the full dataset leads to wrong conclusions on cause and effect

- *"The next Netflix." "The Uber of [fill in the blank of an industry]."* You've likely seen these phrases used in reference to buzzed-about new startups
  - mimicking a successful company or person doesn't guarantee you or your company will achieve the same success
  - many founders try to fit their business into a model that just isn't right for the current market, their audience, or their growth stage

- "Pineapple's *marketing team used these email templates to increase sales by 35%. I'll use these templates and get the same results.*"
  - we must consider the other businesses who tried the same fix and saw lackluster results

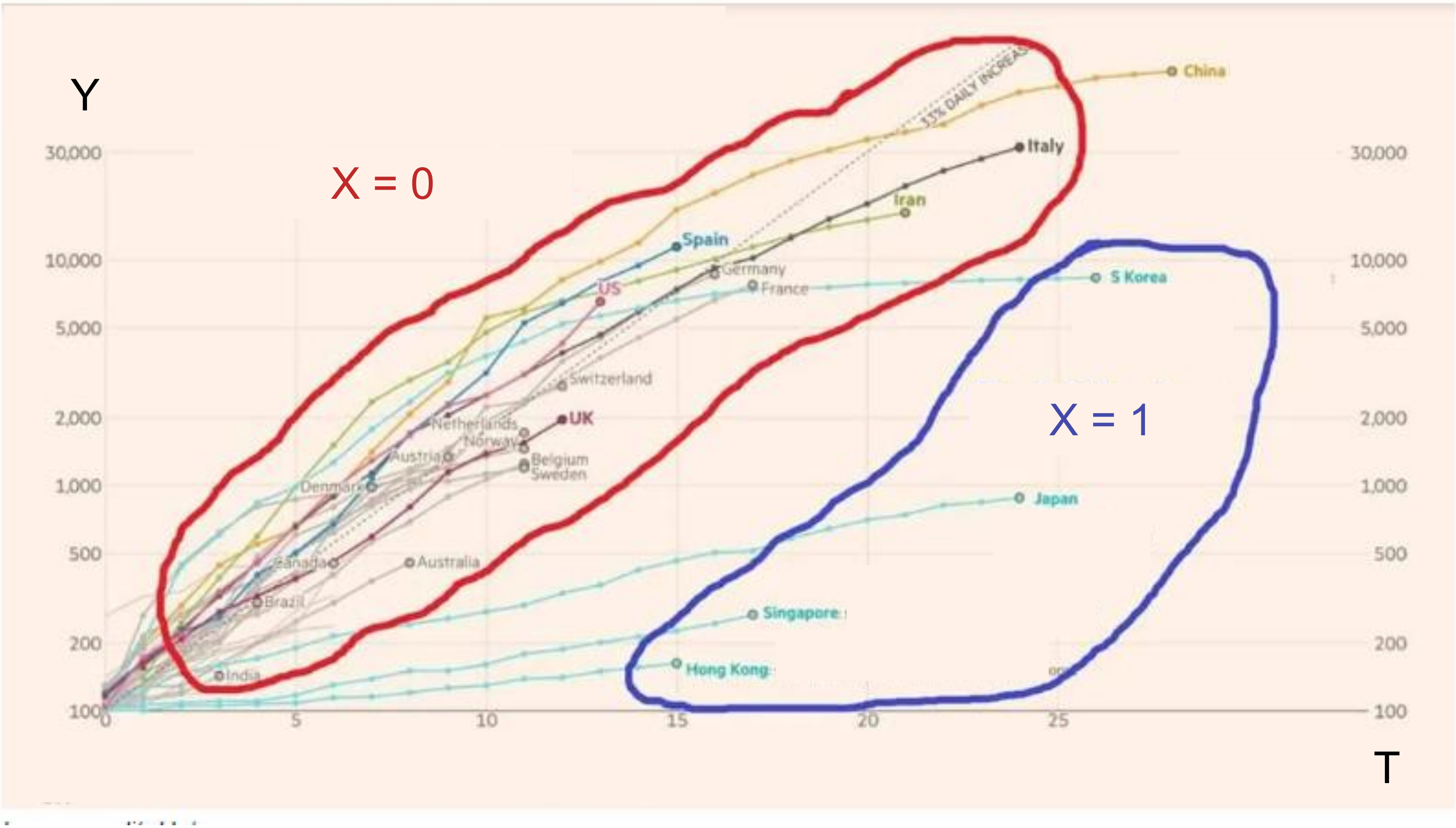
Understand your information/data

Why do people typically have fewer friends than their friends have?



Friendship Paradox: **Sampling bias** in which people with greater numbers of friends have an increased likelihood of being observed among one's own friends.

Linear regression: Y ~ T + X + country



Empirical models

- **What kind of models?**
  - Do you need predictive models? This is easier, just use different machine learning models and have a horse race between the models in out-of-sample.
  - Or do you need to identify causal relationships? This is harder.
  - Or do you need an optimization model?
- Careful **model evaluation** needed
  - Generalization vs. over-fitting
  - Prefer parsimonious model, robustness is important
  - Ensure practical significance as well statistical significance

Predictive model examples (e.g. in other courses)

- Manufacturing:
  Forecasting maintenance by using data from sensors
- Health care:
  Forecasting patients at risk, epidemics
- HR:
  Forecasting employees who are at risk of leaving the company
- Retail industry:
  Forecasting good location for new stores, demand
- Government:
  Forecasting violence
- Finance:
  Forecasting returns and risks, credit scoring

Predictive model examples (Cont'd)

- Forecasting stock market returns and recessions is really difficult
  - E.g. the average alpha of mutual funds is about zero
- How could e.g. regulators do this better if they want to forecast market stress events or use countercyclical capital requirements for banks?

Predictive model examples (Cont'd)

- The first problem (forecasting stock returns) is difficult
- But if we have individual level investment data, then we have a **simple classification problem**:
  - Identify good market timers and follow their behavior
  - We do this by using investors' trades from the Helsinki Stock Exchange

Predictive model examples (Cont'd)

- We find evidence that **some people persistently time the market**
  - Timing occurs at short and long horizons
  - Previously successful timers can expect to outperform
  - Others persistently mistime and underperform

- **Observing timers help policy makers anticipate crashes**
  - The unconditional probability of a bear market is 24.1%, but the probability rises to 60.0% when the good timers' investment flow is at least 0.5 standard deviation below the mean

- In a big data environment, sometimes we need to **rethink the problem** we try to solve

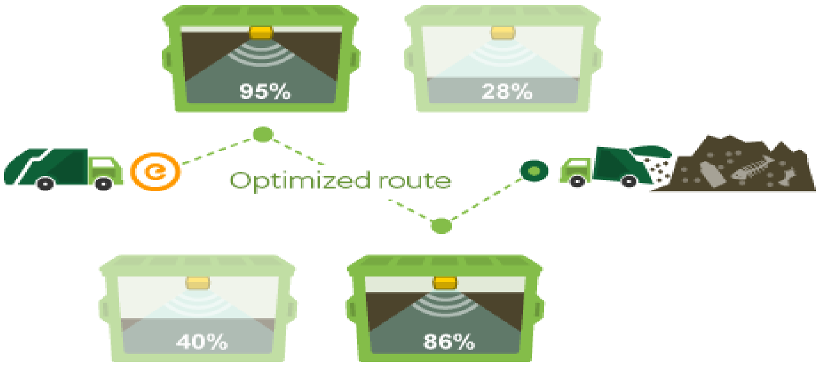
Analytics business example: Waste management and recycling business

- Everyone is familiar with this business, trash collection takes place in many places
  - Quite high competition
  - Over many years, processes have been optimized

- **Can we change the business model in this industry with analytics?**
  - If not, why not?
  - If yes, how?

Sensors, forecasting, optimization…

- Install **wireless sensors** to waste containers
  - These sensors send regularly the fill-level information to business analysts
- Based on the fill-level information **make forecasts** for different days or even hours
- Based on the fill-level information and the forecasts, **optimize collection plan**
  - So, make most efficient schedules and routes by taking into account future fill-level projections, truck availability, traffic information, road restrictions etc.
- **Guide drivers** with a mobile app along the optimal routes and enable them to report problems



Has this been done already?

Yes: OTTO Waste Systems (Singapore), Enevo (Europe)

- Significantly reduces costs, emissions, road wear, vehicle wear, noise pollution, and work hours
  - Provides up to 50% in cost savings
- Can we improve this business model?

Identification of causal relationships (this course)

- Four **identification strategies**:
  - Randomized experiment
  - Regression discontinuity (we might not have time for this)
  - Difference in differences
  - Instrumental variables

- Estimating **causal relationship** is important e.g. when we **change our policy** (e.g. prices)

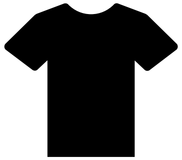
- If the objective is to just forecast (e.g. future returns, volatility, or demand quantity) then we do not need the above identifications
  - Just have a "horse race" between model candidates and check which works the best in out-of-sample

"Deer Lady"

Correlation is not causality

?

Case study



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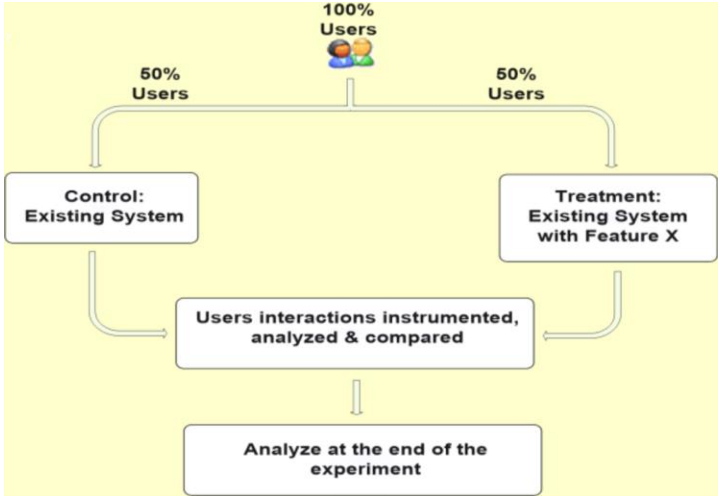
- Correlation: High shirt sales tend to be observed with high shoes sales
- Prediction: If shoe sales are high, shirt sales will be high
- Action: Run promotions on shoe sales to drive shirt sales
- ***What could go wrong?***

Randomized experiment example: A/B Testing

- A/B Testing, also known as bucket testing, split testing, is controlled/randomized experiment in the Internet Age
  - Online controlled experiment
- It is a crucial step in the Lean Startup Method and also heavily used in large online service companies including Amazon, EBay, Facebook, Google, Microsoft, Netflix, etc.
- Like clinical trial for drugs, decisions on website features are heavily based on A/B testing



LinkedIn Marketing Solutions shared:

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Regression discontinuity

- Example: The impact of minimum drinking age on mortality

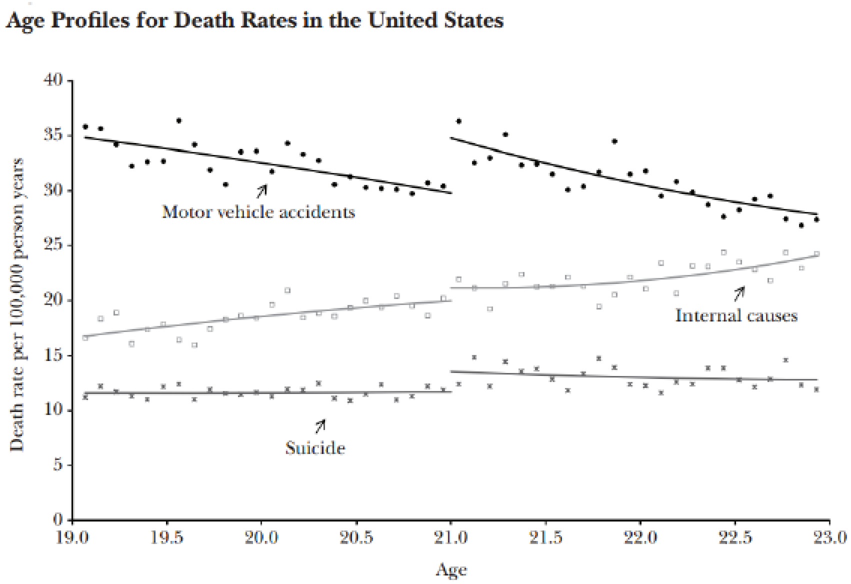
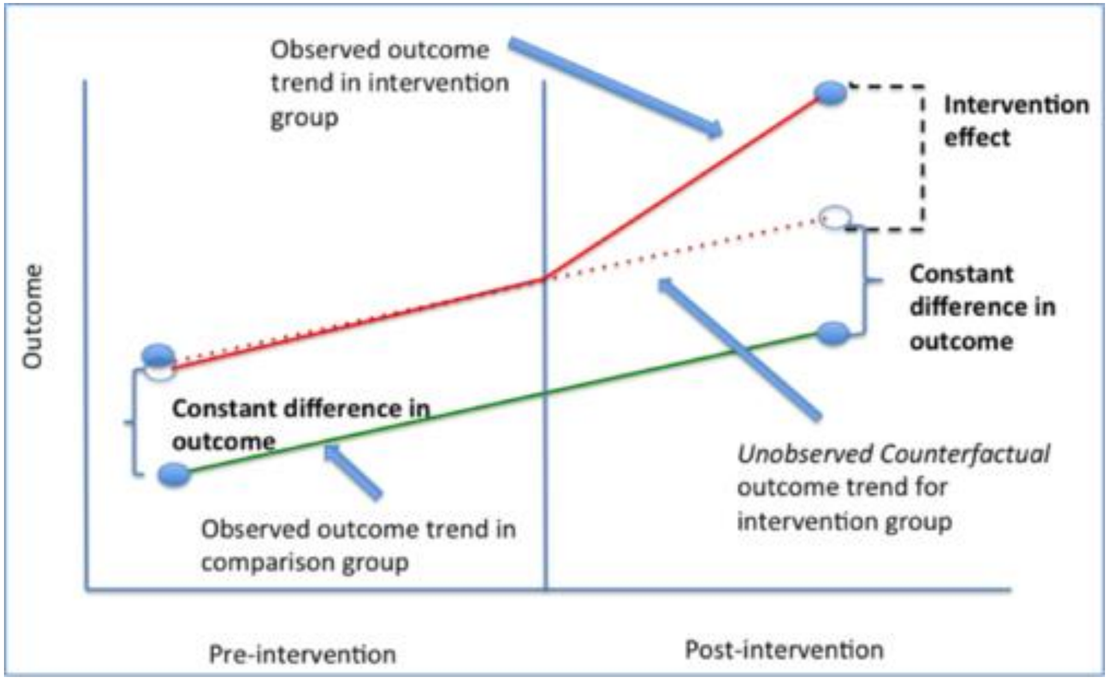


Figure 1: Death rates by age by type.

Difference in differences



Price discrimination

Multi-armed bandits

A Multi-Armed Bandit Framework for Recommendations at Netflix

Jaya Kawale
Elliot Chow

NETFLIX